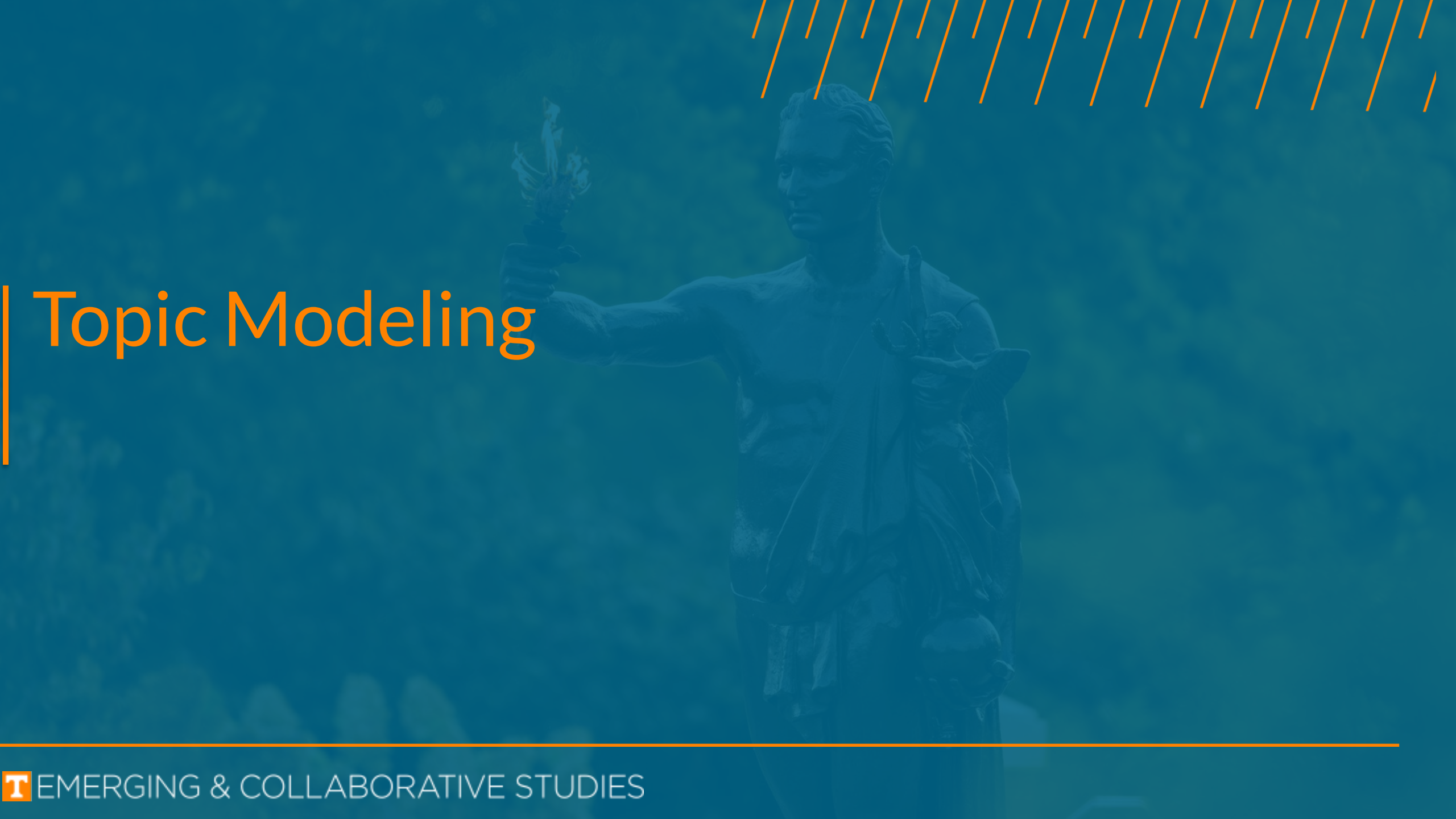

AI-102 Natural Language Processing (NLP)

Dr. Jimmy Ardiansyah



EMERGING &
COLLABORATIVE STUDIES





Topic Modeling

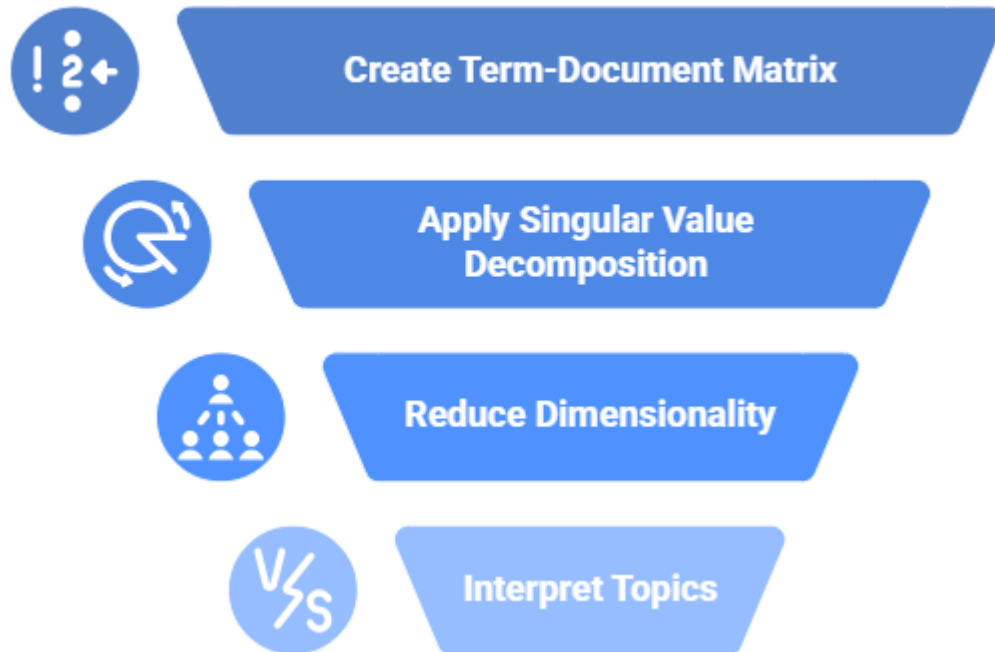
Latent Semantic Analysis (LSA)

Latent Semantic Analysis (LSA) is an important technique in Natural Language Processing (NLP) and Information Retrieval (IR). It is a statistical method for analyzing the relationships between a set of documents and the terms they contain by producing a set of concepts related to the documents and terms.

It does this by turning text into a matrix and then using **Singular Value Decomposition (SVD)** to compress that matrix into a lower dimensional space where underlying topics become more visible



Steps Involved in LSA



Step 1: Term–Document Matrix

- This is the original high dimensional view of the corpus

Step 2: Apply SVD

Decompose the term–document matrix into three matrices:

- U : Term–concept associations (how words load onto latent concepts)
- Σ (Sigma): Diagonal matrix of singular values and Shows how important each concept is
- V^T : Document–concept associations (how documents load onto latent concepts)

Step 3: Dimensionality Reduction

- Keep only the top k singular values and corresponding columns of U and rows of V^T
- Drops small singular values that represent noise

Step 4: Interpreting Topics

- Each concept/topic is a weighted combination of terms
- Look at the top weighted terms in each component to name the topic

Common LSA Applications

By transforming documents and terms into this reduced conceptual space, LSA enables powerful applications:

- **Information Retrieval:** Users can search using terms that may not appear in the most relevant document, but which belong to the same latent concept.
- **Document Clustering/Classification:** Documents are grouped based on the concepts they contain, making for more semantically meaningful clusters.
- **Word Similarity:** The proximity of words in the dimensional space can be used as a measure of their semantic similarity.



Latent Dirichlet Allocation (LDA)

Latent Dirichlet Allocation is a **probabilistic topic modeling method** that represents each document as a **mixture of topics**, and each topic as a **mixture of words with certain probabilities**. It assumes a generative process: for every word in a document, the model first picks a topic (based on the document's topic distribution), then picks a word (based on that topic's word distribution). By fitting this model to a corpus, LDA uncovers the **hidden thematic structure** in large collections of text.

- Each **document** is a mixture of topics.
- Each **topic** is a mixture of words (with probabilities).

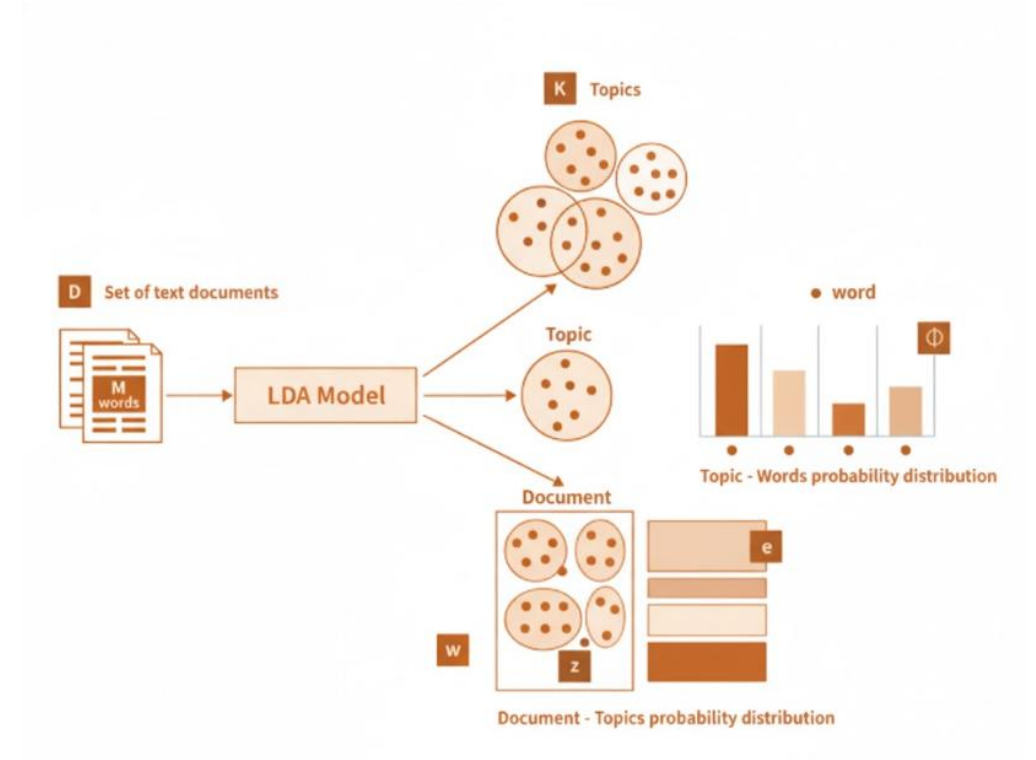


"I've got to knock out the paintings. AI needs more training materials."

Understanding Latent Dirichlet Allocation (LDA)

Topic Distribution (per document)

- Each document has a probability over topics.
- Example: 40% Environment, 30% Politics, 20% Economics, 10% Health



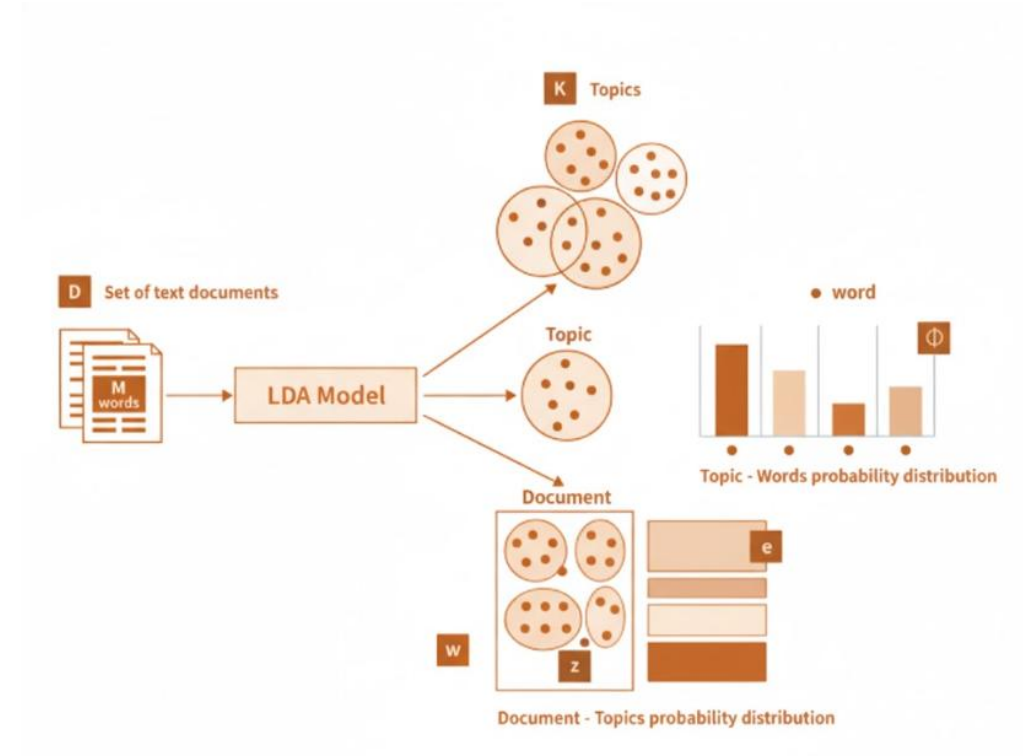
Understanding Latent Dirichlet Allocation (LDA)

Topic Distribution (per document)

- Each document has a probability over topics.
- Example: 40% Environment, 30% Politics, 20% Economics, 10% Health

Word Distribution (per topic)

- Each topic has a probability over words.
- Sports topic: “game”, “team”, “score”, “player” likely words.



Understanding Latent Dirichlet Allocation (LDA)

Topic Distribution (per document)

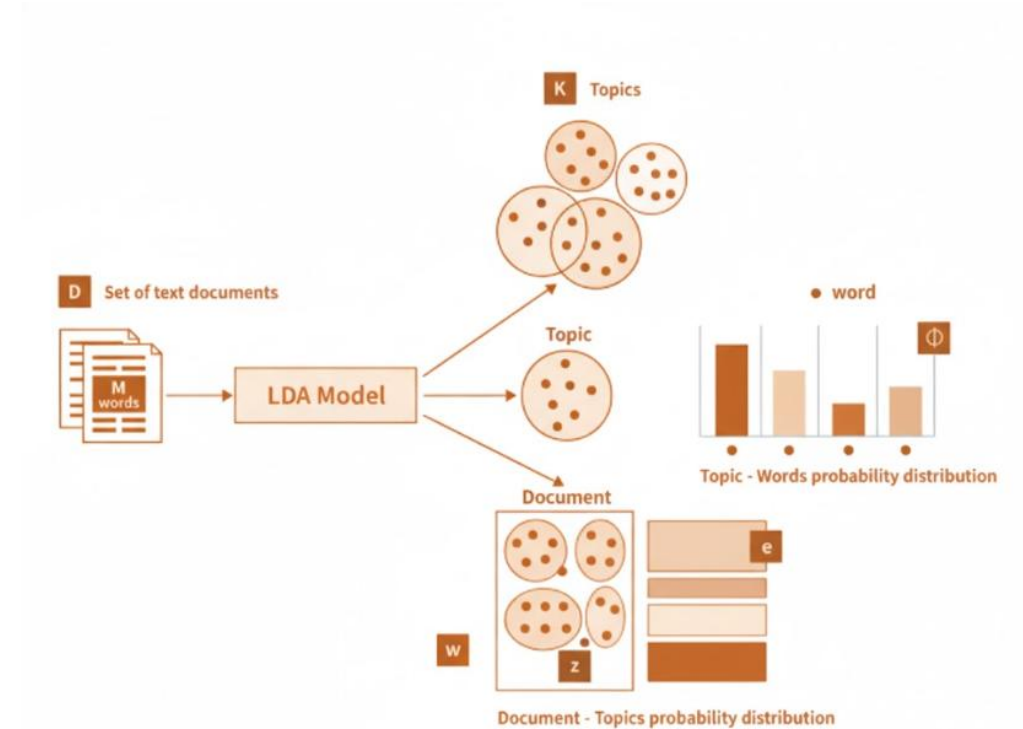
- Each document has a probability over topics.
- Example: 40% Environment, 30% Politics, 20% Economics, 10% Health

Word Distribution (per topic)

- Each topic has a probability over words.
- Sports topic: “game”, “team”, “score”, “player” likely words.

Generating a Document

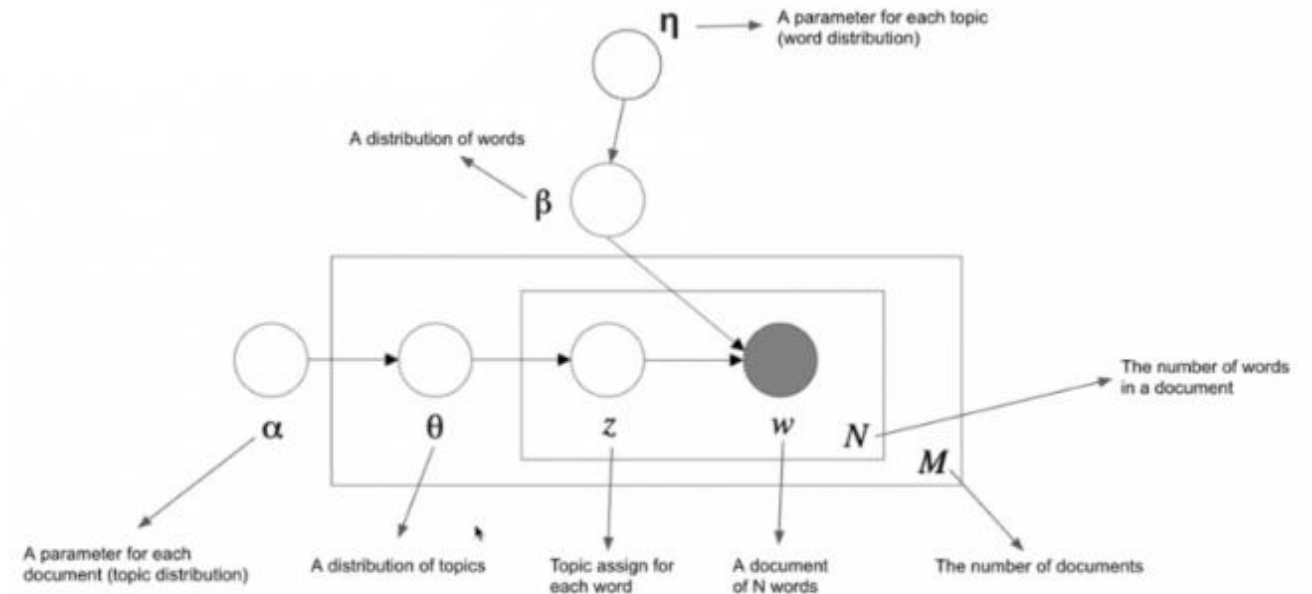
- Select a topic for the current word
- Select a word given that topic
- Climate change example: one word might come from “Environmental Science” (“emissions”), the next from “Politics” (“policy”).



Mathematical Formulation of LDA

All of them together let LDA **uncover hidden thematic structure** in a corpus in a principled probabilistic way.

- Dirichlet priors $\rightarrow$ make θ and β sparse.
- β (topic-word distributions) $\rightarrow$ what each topic “talks about”.
- θ (document-topic distributions) $\rightarrow$ what mix of topics each document has.
- z (word assignments) $\rightarrow$ which topic generated each word.



Hierarchical Dirichlet Process (HDP)

Hierarchical Dirichlet Process is a nonparametric extension of LDA that lets the model discover the number of topics automatically

Imagine you have a collection of documents, and these documents are naturally grouped—for example, by the newspaper they appeared in (Group 1: *The New York Times*, Group 2: *The Wall Street Journal*, etc.).

- Each newspaper (group) will have its own set of topics (clusters), like "Politics" or "Finance."
- Some topics will be unique to one paper, but many will be **shared** across all papers (like "Sports" or "Weather").

HDP is designed to model this situation:

- It treats each group (newspaper) as having its own mixture of topics.
- Critically, it forces all groups to draw their topics from a single, shared, master list of all possible topics. This is the **Hierarchical** part.



Hierarchical Structure of HDP

The most famous way to understand HDP is through the **Chinese Restaurant Franchise** analogy, which extends the simpler Dirichlet Process's "Chinese Restaurant Process" (CRP) analogy:



Hierarchical Structure of HDP

The most famous way to understand HDP is through the **Chinese Restaurant Franchise** analogy, which extends the simpler Dirichlet Process's "Chinese Restaurant Process" (CRP) analogy:

Concept	CRF Analogy	HDP Meaning
The Franchise	A Chain of Restaurants	All of your Groups of Data (e.g., all the newspapers).
The Global Menu (G_0)	A Master Menu with an infinite list of dishes (topics).	The Global Distribution of Topics .
A Single Restaurant (G_j)	A Single Restaurant in the chain.	A single Group of Data (e.g., one newspaper).
Dishes Served	The Dishes actually served at that restaurant.	The Topics present in that one document group.
Customer (x_{ji})	A Customer who walks into the restaurant.	A single Data Point (e.g., a word in a document).

Hierarchical Structure of HDP

The most famous way to understand HDP is through the **Chinese Restaurant Franchise** analogy, which extends the simpler Dirichlet Process's "Chinese Restaurant Process" (CRP) analogy:

Concept	CRF Analogy	HDP Meaning
The Franchise	A Chain of Restaurants	All of your Groups of Data (e.g., all the newspapers).
The Global Menu (G_0)	A Master Menu with an infinite list of dishes (topics).	The Global Distribution of Topics .
A Single Restaurant (G_j)	A Single Restaurant in the chain.	A single Group of Data (e.g., one newspaper).
Dishes Served	The Dishes actually served at that restaurant.	The Topics present in that one document group.
Customer (x_{ji})	A Customer who walks into the restaurant.	A single Data Point (e.g., a word in a document).

How the Process Works:

1. The Franchise Owner's Decision (Top Level): The master chef (the top-level DP) decides on an infinite, shared Master Menu of all possible dishes (topics).
2. The Local Chef's Decision (Group Level): When a new restaurant opens, the local chef (the group-level DP) looks at the Master Menu. They choose a subset of those dishes for their local menu, but they cannot invent a new dish that isn't on the Master Menu.
3. The Customer's Decision (Data Level): A customer (a word) walks into a restaurant (a document group) and sits at a table (topic).
 - They can sit at an occupied table (assign to an existing topic).
 - They can start a new table (introduce a new topic to that group).
 - If they start a new table, the dish served at that table must be one of the shared dishes from the Master Menu.





EMERGING &
COLLABORATIVE
STUDIES
